

CSM-322: Information and Coding Theory

Lab - 7

(Odd Semester 2024-25)

October 24, 2024

Q1. Syndrome Decoding Rule

For given a binary linear code C , write a program to construct the syndrome look-up table considering complete decoding scheme. Use the syndrome look-up table constructed to decode the received word w .

Example: $C = \{0000, 1011, 0101, 1110\}$, $w = 1101$

Output:

$G =$

1 0 1 1

0 1 0 1

$H =$

1 0 1 0

1 1 0 1

Syndrome Table: {

00: 0000,

11: 1000,

01: 0100,

10: 0010

}

Sent word: 0101